

Name of the Programme:	B.Tech(Hons.)	
Title of the Course:	Linear Algebra	
Course code	CS1807	
Semester:	II	
Names of the Course Instructor(s)	Dr. Shrikant Utagi Dr. Rekha Satish Dr. Manjunath S Prof. Sasikala J.	
Course credit	No. of hours per week (2+1+0)	Total no. of Teaching hours
03	03 Hours	45 Hours

Session	Module No.	Topic	Pedagogy /activities	Pre-reading/ Reference
1	1	System of linear equations; row echelon form;	Lecture + PPT/Activity	Poole, Ch-2, P63-140
2	1	consistent and inconsistent equations;	Lecture + chalk and Talk/ PPT/Activity	Poole, Ch-2, P63-140
3	1	Gaussian elimination;	Lecture + chalk and Talk/ PPT/Activity	Poole, Ch-2, P63-140
4	1	Tutorial: Gaussian elimination;	Discussion + Examples	Poole, Ch-2, P63-140
5	1	reduced row echelon form;	Lecture + chalk and Talk/ PPT/Activity	Poole, Ch-2, P63-140
6	1	Gauss-Jordan Elimination, Inverse;	Lecture + chalk and Talk/ PPT/Activity	Poole, Ch-3, P142-262
7	1	Tutorial: Gauss-Jordan Elimination, Inverse;	Discussion + Examples	Poole, Ch-3, P142-262
8	1	LU decomposition; Gauss-Seidel method.	Lecture + chalk and Talk/ PPT/Activity	Strang, Ch2, P31-109, Poole, Ch-3, P142-262

9	1	Tutorial: LU decomposition; Gauss-Seidel method.	Discussion + Examples	Strang, Ch2, P31-109, Poole, Ch-3, P142-262
10	2	Homogeneous equations;	Lecture + chalk and Talk/ PPT/Activity	Poole, Ch-6, P536
11	2	Homogeneous equations (continued);	Lecture + chalk and Talk/ PPT/Activity	Poole, Ch-6, P536
12	2	Tutorial: Homogeneous equations;	Discussion + Examples	Poole, Ch-6, P536
13	2	span; linear independence/dependence of basis;	Lecture + chalk and Talk/ PPT/Activity	Strang, Ch3, P123-181,
14	2	span; linear independence/dependence of basis (continued);	Lecture + chalk and Talk/ PPT/Activity	Strang, Ch3, P123-181,
15	2	Tutorial: span; linear independence/dependence of basis;	Discussion + Examples	Strang, Ch3, P123-181,
16	2	subspaces; dimension, rank and nullity;	Lecture + chalk and Talk/ PPT/Activity///	Strang, Ch3, P123-181
17	2	rank-nullity theorem (continued)	Lecture + chalk and Talk/ PPT/Activity	Strang, Ch3, P123-181
18	2	Tutorial: subspaces; dimension, rank and nullity; rank-nullity theorem	Discussion + Examples/Activity/ /	Strang, Ch3, P123-181;
19	3	Linear transformations, domains; codomains; image;	Lecture + chalk and Talk/ PPT/Activity	Strang, Ch8, P401-421
20	3	Linear transformations, domains; codomains; image; (continued)	Lecture + chalk and Talk/ PPT/Activity	Strang, Ch8, P401-421
21	3	Tutorial: Linear transformations, domains; codomains; image;	Discussion + Examples/Activity/	Strang, Ch8, P401-421

22	3	range and kernel of a linear map; composition of linear maps;	Lecture + chalk and Talk/ PPT/Activity	Strang, Ch8, P401-421
23	3	Tutorial: range and kernel of a linear map; composition of linear maps;	Discussion + ExamplesActivity/ /	Strang, Ch8, P401-421
24	3	eigenvalue equation; Characteristic equation;	Lecture + chalk and Talk/ PPT/Activity	Strang, Ch6, P288-350
25	3	eigenvalue and eigenvectors;eigenspaces;	Lecture + chalk and Talk/ PPT/Activity	Strang, Ch6, P288-350
26	3	similarity and diagonalization.	Lecture + chalk and Talk/ PPT/Activity	Strang, Ch6, P288-350
27	3	Tutorial: eigenvalues, eigenvectors, characteristic equation, similarity and diagonalization;	Discussion + ExamplesActivity/ /	Strang, Ch6, P288-350; Strang, Ch6, P288-350
28	4	Orthogonality, Orthonormality, Basis set	Lecture + chalk and Talk/ PPT/Activity	Strang, Ch4, P194-233; Poole, Ch-5, P377-443
29	4	orthogonal and orthonormal: -basis, matrices, projections, decomposition;	Lecture + chalk and Talk/ PPT/Activity	Strang, Ch4, P194-233; Poole, Ch-5, P377-443
30	4	Tutorial: orthogonal and orthonormal: -basis, matrices, projections, decomposition;	Discussion + Examples/Activity/ /	Strang, Ch4, P194-233; Poole, Ch-5, P377-443
31	4	Gram-Schmidt orthogonalization;	Lecture + chalk and Talk/ PPT/Activity	Strang, Ch4, P194-233; Poole, Ch-5, P377-443
32	4	Gram-Schmidt orthogonalization (continued);	Lecture + chalk and Talk/ PPT/Activity	Poole, Ch-5, P377-443
33	4	Tutorial: Gram-Schmidt orthogonalization	Discussion + Examples/ Activity/	Strang, Ch4, P194-233; Poole, Ch-5, P377-443

34	4	QR decomposition; orthogonal diagonalization;	Lecture + chalk and Talk/ PPT/Activity	Poole, Ch-5, P377-443
35	4	QR decomposition; orthogonal diagonalization (continued)	Lecture + chalk and Talk/ PPT/Activity	Poole, Ch-5, P377-443
36	4	Tutorial: QR decomposition; orthogonal diagonalization	Discussion + Examples/Activity	
37	5	Spectral theory;	Lecture + chalk and Talk/ PPT/Activity	Poole, Ch-4, P419-443
38	5	quadratic forms;	Lecture + chalk and Talk/ PPT/Activity	Poole, Ch-5, P419-443
39	5	Tutorial: Spectral theory and quadratic forms;	Discussion + Examples/Activity	Poole, Ch-4, P419-443; Poole, Ch-5, P419-443
40	5	principal axis theorem; norm and distance;	Lecture + chalk and Talk/ PPT/Activity	Poole, Ch-5, P419-443
42	5	least squares approximation;	Lecture + chalk and Talk/ PPT/Activity	Strang, Ch4, P219; Poole, Ch-7, P591
43	5	Tutorial: norm and distance; least squares approximation;	Discussion + Examples/Activity	Poole, Ch-5, P419-443; Strang, Ch4, P219; Poole, Ch-7, P591
43	5	Singular Value Decomposition	Lecture + chalk and Talk/ PPT/Activity	Strang, Ch8, P364-392; Poole, Ch-7, P613-633
44	5	Singular Value Decomposition (Continued)	Lecture + chalk and Talk/ PPT/Activity	Strang, Ch8, P364-392; Poole, Ch-7, P613-633

45	5	Tutorial: Singular Value Decomposition	Discussion + Examples/Activity	Strang, Ch8, P364-392; Poole, Ch-7, P613-633
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Assessment and Evaluation

Continuous Internal Evaluation (CIE) :70%	
Components of CIE	Weightage of each component (Marks)
- CIE 1 (Pen and Paper)	20
- CIE 2 (Pen and Paper)	25
- CIE 3-A (Quiz via Google form/Tutorial)	5
- CIE 3-B (Application-based project in groups of 4 or 5 each)	20

Semester End Exams (SEE)	
Mode of Exam	Pen and Paper
Weightage of exam	30

Course Outcomes- Programme Outcomes Matrix

Program Outcome Course Outcome		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	Apply the concepts of linear algebra to solve a given set of linear equations	3	2			2			2		2
CO2	Apply concepts of rank, nullity to problems of linear algebra	3	2			2			2		2
CO3	Apply the concepts of linear transformation theory to solve eigenvalue problems	3	2		2	2			2		2
CO4	Apply principles of orthonormality,	3	2		2	2			2		2

	spectral theory and singular value decomposition to applied linear algebra problems										
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Rubrics for evaluation of CIE (separate for each CIE component)

Total Marks		20
CIE-1 Components		Rubrics for Assessment
CIE-1	Pen and Paper Exam	The test will be conducted for 20 marks and will be evaluated according to the Scheme of Evaluation prepared by the course faculty.

Total Marks		25
CIE-2 Components		Rubrics for Assessment
CIE-2	Pen and Paper Exam	The test will be conducted for 25 marks (Pen and Paper) and will be evaluated according to the Scheme of Evaluation prepared by a team of course faculty.

Total Marks		25
CIE-3 Components		Rubrics for Assessment
CIE-3-A	Quiz via Google form / Tutorial	Quiz or Tutorial will be conducted for 10 marks (will be scaled down to 5 marks)
CIE-3-B	Application-based project	Activities will be conducted for 20 marks. (Refer TABLE-1)

TABLE-1 : Graded Component CIE-3-B (20 Marks):

Mini Project: Applications of Linear Algebra Rubrics (Max: 20 M)					
1	Content Accuracy & Application of Linear Algebra (8 Marks)	Content	All linear algebra concepts are correct, well applied, and logically justified. (7–8 Marks)	Most answers are correct; minor errors present. (4-6 Marks)	Several incorrect answers; lacks understanding (0–3 Mark)

2	Presentation Quality & Communication (6 Marks)	Presentation	Clear, well-structured presentation; effective visuals; confident communication (5–6 Marks)	Moderately organized presentation; explanation partially clear. (3–4 Marks)	Poor organization; unclear explanation; weak presentation. (0–2 Marks)
3	viva (6 Marks)	Viva Voce	Excellent understanding and confident explanation; uses Octave effectively. (5–6 Marks)	Moderate understanding; explains partially with some guidance. (3–4 Marks)	Poor or no understanding ; unable to answer or use Octave. (0–2 Marks)

Signature of the Course Faculty